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A LOOK AT PROCESSORS

Before we get to the action, let us refresh our knowledge on the types of processors available.

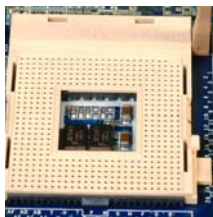
Intel Processors

As of today, Intel has two lines of processors, namely the Pentium 4 series for performance seekers, and Celeron D for the value-end of the market. The original Pentium 4 processors were based on the 130nm manufacturing process, featured 512 KB of L2 cache and ran on a 533 MHz FSB. But today's Pentium 4 is a different beast; it features 1 MB of L2 cache, runs on an 800 MHz FSB and is manufactured using the 90nm process. A still newer version expected soon will run on a 1066 MHz bus.

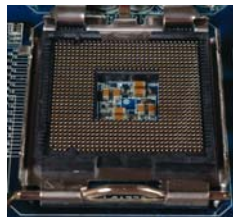
There are also special-edition Pentium 4 processors called 'Extreme Edition', and denoted by 'EE'. The EE processors have 2 MB

of L3 cache and are manufactured using the older 130nm process. The new line of Pentium 4 processors based on the LGA 775 package is denoted by a new nomenclature. Instead of using the core speed as the name, Intel calls this series the 5XX series, where the 'XX' ranges from 20 to 60. Processors that have the 'Execute disable bit' capability have the letter 'J' following the 5XX number, an example being 560J.

The older Pentium 4 processors required a Socket 478 motherboard, e.g. 845, 865, 875, chipset-based boards, whereas the newer ones will be produced for the LGA 775 package. The LGA



The older FC-PGA 478 interface



The new LGA 775 interface

(Land Grid Array) type of socket is found on motherboards supporting Intel's new chipsets, i.e., the 915 and 925.

On the Celeron front, too, there are two versions; the older ones that run on a 400 MHz bus have 128 KB of L2 cache, and are manufactured using the 130nm process. The newer 90nm-manufactured Celeron processors are called Celeron D, and are denoted by 3XX series where the 'XX' ranges from 15 to 40. The new Celeron D runs on a 533 MHz bus and features 256 KB of L2 cache. Some models of the Celeron D are available for the LGA 775 sockets; the rest use the older Socket 478.

AMD Processors

AMD has the Athlon 64 and Sempron ranges of processors. Depending on the processor interface, or the socket, the Athlon 64 can be divided into two distinct categories—one that uses Socket 754 and the other that plugs into a Socket 939. 754-pin processors

How We Tested Processors

This year, we revived our 'Processor Test' process with newer and better benchmarks and testing procedures. All the new benchmarks are highly processor-dependent, and the peripheral sub-system has a low level of involvement, which results in scores that reflect the true potential of the processor's computational power.

For Intel's LGA 775 processors, we used Intel's D952XCV motherboard with 1 GB of DDR-II 533 MHz memory, whereas for the Socket 478 processors, we used Gigabyte's 8TR5350MT motherboard with 1 GB of DDR 400 MHz Corsair memory.

AMD's Socket 939 processors were plugged onto an Asus A8V-Deluxe, while Socket 754 processors went with MSI's K8N Neo 2. Socket 'A' processors were tied with the ASUS A7N8X-Deluxe.

Windows XP with SP1 was freshly loaded on a Maxtor Diamond Max 120 GB SATA drive. On the graphics front, we used Gigabyte's AGP card based on the nVidia GeForce 5700 chipset and MSI's PCIe card based on nVidia's GeForce 5750 chipset.

ScienceMark 2.0

ScienceMark 2.0, as the name suggests, is a benchmark based on scientific calculations. This benchmark constitutes multiple benchmarks that test various aspects of a processor. We used the following benchmarking modules:

MolDyn (Molecular Dynamics): Molecular Dynamics simulates the thermodynamic behaviour of materials using their forces, velocities, and positions, where force is the most important parameter.

Primordia (atoms): This code calculates the Quantum Mechanical Hartree-Fock orbitals for each electron in any element of the periodic table. The number crunching involved in the calculation is complex and stresses the FPU (Floating-Point Unit) of a processor. The final score is given as the time taken to complete the calculations.

Cipher Test: The 'Cipher Test' tests four different forms of encryption algorithms, AES 128/256 bit and RSA 512/1024 bit. The calculations involved are mathematically inclined, resulting in the stressing of the ALU (Arithmetic and Logic Unit) of the processor.

WinRAR 3.3

WinRAR 3.3 offers an inbuilt benchmark, operating on random data, which is processed in the memory. Since the data is operated on in the memory, the results are free from any influence that a disk system may impart.

CINEBENCH 2003

CINEBENCH 2003 is the free benchmarking tool for Windows and Mac OS based on the 3D software CINEMA 4D R8. CINEBENCH 2003 delivers accurate results by testing a processor's raw processing speed which affects system performance such as OpenGL, multithreading and multiprocessing. Also, this benchmark gives a clear view of Intel's HT technology in operations, as it supports HT.

SiSoft Sandra 2005

This is the same good old benchmarking tool with newer features and tests added to it. Nothing seems to have changed except for the new user interface and the extra benchmarking tool for USB drives, cache memory and other parts. We logged scores particularly related to the CPU, memory and disk drive.

Doom 3

Doom 3 is a graphically intensive game capable of making any machine crawl. This OpenGL game paints a good picture of what a decent machine should consist of to run games of the future.

Half-Life 2

This DirectX 9.0 game scales very well with processor speed. Moreover, any hardware which runs this game well will be comfortable with upcoming game titles.

Ziff-Davis Multimedia Content Creation

This benchmark suite runs applications related to processor-intensive multimedia applications such as Adobe Photoshop, Adobe Premier and WaveLab.